

Andrés Riveros Valdevenito

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Education

PhD	Columbia University , PhD in Statistics. • Co-advised by Professor Marcel Nutz & Professor Philip Protter.	Sep 2020 - May 2026
M Phil	Columbia University , Master of Philosophy in Statistics. • As part of the PhD program.	Sep 2020 - May 2024
MA	Columbia University , Master of Arts in Statistics. • As part of the PhD program.	Sep 2020 - Feb 2023
MS	Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile , Master of Engineering Sciences, minor in Applied Mathematics. • With Maximum Distinction. • Advised by Professor Jaime San Martín.	Mar 2014 - Jun 2020
Prof. Deg.	Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile , Mathematical Engineer. • With Maximum Distinction.	Mar 2014 - Jun 2020
BS	Facultad de Ciencias Físicas y Matemáticas (FCFM), Universidad de Chile , Engineering Sciences in Mathematics. • With Distinction.	Mar 2014 - Dec 2018

Teaching

Teaching Assistant at Columbia University :	Sep 2020 - May 2026
• Stochastic Methods in Finance STAT GR4265 (Fall 2025), • Probability Qualifying Exam Review Session Teacher (Summer 2023, 2024, 2025), • PhD Probability II STAT GR6302 (Spring 2024, Spring 2025, Spring 2026), • PhD Probability I STAT GR6301 (Fall 2023, Fall 2024), • Core Competency Exam Review Session Teacher (Spring 2024), • Probability STAT S4203/W5203 (Summer 2023), • Introduction to Statistics with Calculus STAT S1201/W1201 (Summer 2022, Spring 2023), • Linear Regression Models STAT S4205/S5205 (Fall 2022), • Introduction to Statistics STAT W1101 (Spring 2022, Fall & Spring 2021), • Introduction to Statistical Reasoning STAT W1001 (Summer 2021, Fall 2020).	
Teaching Assistant at Universidad de Chile, FCFM:	Mar 2014 - Jun 2020
• Supervisor of Entire Core Courses of Mathematics (2019,2020,2021), • Stochastic Calculus MA5402 (Spring 2018, 2019), • Markov Processes MA4401 (Fall 2018), • Introduction to Calculus MA1001 (Fall 2017), • Linear Algebra MA1102 (Spring 2016): Grader & Test Supervisor, • Introduction to Calculus MA1001 (Fall 2016): Grader & Test Supervisor.	

Honours & Awards

Columbia University:	Sep 2020
• Dean's Fellow, Graduate School of Arts & Sciences.	
ANID (CONICYT):	Mar 2019
• Excellence Scholarship for National Science Masters Degree Students, Chile.	

Experience

Dept. of Applied Mathematics, Universidad de Granada, Visiting Researcher.	Granada, Spain
• Invited by Professor Alejandro Gárriz.	Jun 2025
• Ongoing project related to optimal transport theory.	
Center of Mathematical Modelling, Universidad de Chile, Research Intern.	Santiago, Chile
• Under guidance of Professor Joaquín Fontbona.	Jan 2019
• Research review of state-of-the-art HMC methods.	
Superintendency of Banks and Financial Institutions (SBIF, now Financial Market Commission, CMF), Analytics Intern.	Santiago, Chile
• Under Patricio Jaramillo, director of the area of modelling and methodology.	Jan 2018
Assarlab, Faculty of Medicine, Universidad de Chile, Research Intern.	Santiago, Chile
• Under professor Rodrigo Assar.	Jan 2017
• Created algorithms based on previous research to analyze the learning process of fish.	

Publications

Linear Convergence of Gradient Descent for Quadratically Regularized Optimal Transport.	2026+
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , Preprint, arXiv:2509.08547  .	
Polyak-Łojasiewicz Inequality for Quadratically Regularized Optimal Transport	2026
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , SIAM Journal on Optimization (To appear), arXiv:2605.27175  .	
Monotonicity in Quadratically Regularized Linear Programs.	2025
A. González-Sanz, M. Nutz, A. Riveros Valdevenito , SIAM Journal on Optimization, Vol 35, N°2, pp 1419-1437, arXiv:2408.07871  .	
Markov Jump Times and their Cox Construction.	2024
P. Protter, A. Riveros Valdevenito , Annals of Operations Research, arXiv:2310.07003  .	
On the Guyon-Lekeufack Volatility Model.	2024
M. Nutz, A. Riveros Valdevenito , Finance & Stochastics, Vol 28, N°4, pp 1203-1223, arXiv:2307.01319  .	
Propiedad de Martingala, Volatilidad Estocástica & Burbujas Financieras	2020
A. Riveros Valdevenito , Master's Thesis, advised by professor Jaime San Martín, available at respositorio uchile  .	

Invited & Contributing Talks

- *Minghui Conference*, Columbia University, April 2026.
- *IV Workshop on Stochastic Analysis and Optimization, applied to economics, finance, energy and insurance*, Puerto Natales, Chile, December 2025.
- *9th Eastern Conference in Mathematical Finance* (Poster), CMU, October 2025.
- *Optimal Transport Reading Group*, CMU, October 2025.
- *Probability Seminar*, FCFM, Universidad de Chile, August 2025.
- *Young Researchers Seminar*, Universidad de Granada, June 2025.
- *Minghui Conference*, Columbia University, April 2025.
- *Statistics and Optimal Transport Workshop* (Poster), Columbia University, March 2025.
- *Statistics Student Seminar*, Columbia University, January 2025.
- *23rd Northeastern Probability Seminar*, Columbia University, November 2024.
- *Minghui Conference*, Columbia University, April 2024.
- *22nd Northeastern Probability Seminar*, NYU, November 2023.
- *Minghui Conference*, Columbia University, April 2023.
- *Oxford Statistics Student Seminar*, June 2020.

Referee

- Finance & Stochastics.

Technologies & Skills

Basic Programming: Matlab, SciLab, Python, R, Java, Unix, SQL, Visual Basic, Latex, MS Office.

Mathematical Programs: AMPL, BOCOP, FreeFem, STATA, Stan.

Languages: Fluent in English and Spanish. Basic French.